Wireless and Cellular Communication System [18EC81]

[SIMP] — TIE ECE Review team

Module - 01

01. Explain the relating power to electric field used in Mobile Radio Propagation.

02. Explain in detail Three Basic Propagation mechanisms used in mobile radio propagation.

03. Explain Rayleigh Fading and the Line-of-Sight Channels - Ricean distribution in wireless cellular system.

04. Define.

(i) Delay spread and Coherence Bandwidth

(ii) Doppler spread. and Coherence time

(iii) Cell sectoring.

05. Explain three statistical channel model of a broadband fading channel.

06. Problems.

Scanned with CamScanner

Module - 02

01. Explain with diagram GSM n/w architecture (or) explain three major subsystems of a GSM n/w.

02. Draw a neat diagram for the following:
   (a) TDMA hyper frame structure
   (b) GSM air interface time slot.
   (c) TDMA timing offset b/n uplink & downlink.
   (d) GSM traffic and control bursts.

03. With neat diagram explain the following:
   (a) GSM n/w interface
   (b) GSM protocols and signaling model.
   (c) GSM speech processing.

04. Define MSRN. Explain GSM call set up using MSRN. List out 10 GSM operation in call setup and explain with a neat diagram.

05. Explain the logical channel concept or explain various logical channels used in GSM.

Scanned with CamScanner

Module - 03

01. With neat diagram explain the basic spreading procedure used on the forward CDMA channels.

02. Explain the generation of CDMA in
(i) Forward direction
(a) Pilot channel signal
(b) Synchronous channel signal
(c) Paging channel signal
(d) Traffic / Power control channels

(ii) Reverse direction
(a) Access channels
(b) Traffic / Power control channels.

03. Explain CDMA mobile station initialization with a neat flow chart also explain 3-types of CDMA power control techniques.

04. Explain the following:
(a) Idle / Access Handoff
(b) Soft Handoff
(or)
(c) 3-types of soft-CDMA handoff.
(d) W-CDMA and UMTS in 3G CDMA.

05. List number of states in CDMA cell establishment. With neat diagram, explain access channel.

06. With the help of timeline chart, explain
(a) Mobile-originated call
(b) Mobile-terminated call.

Scanned with CamScanner

Module - 04

01. With neat diagram, explain IP based flat-network architecture and explain the elegant approach to intersymbol interference.

02. Explain the advantages of OFDM that led to its selection for LTE. Explain OFDM baseband to passband transmitter.

03. Explain SC-FDE system using comparison between an OFDM system and SC-FDE system. Compare SC-FDE performance v/s OFDM.

04. What is PAR problem? Discuss in detail with the help of amplifier response and PAPR v/s efficiency graph.

05. Explain key steps in an OFDM communication system using vector notation.

06. Discuss circular convolution and DFT for removing ISE in LTE.

Scanned with CamScanner

Module - 05

01. With a neat block-diagram explain the following:

(a) OFDMA downlink transmitter
(b) OFDMA uplink receiver
(c) SC-FDMA uplink transmitter
(d) SC-FDMA uplink receiver.

02. With a neat diagram, explain the following:

(a) Downlink OFDMA resource grid
(b) Uplink SC-FDMA resource grid
(c) LTE radio interface protocol.
(d) LTE N/W architecture.

03. Discuss in detail OFDMA system design considerations.

04. Explain Hierarchial channel structure of LTE.

05. Explain in detail the LTE basic design principles.

06. Draw for LTE downlink

(a) Frame structure type 1
(b) Frame structure type 2

Scanned with CamScanner